

Topic 2: Directory Management			
10	ls	List files and directories	ls
11	ls -l	Display detailed file information	ls -l
12	ls -a	Display all files including hidden files	ls -a
13	cd	Change the current directory	cd Documents
14	cd ..	Move to the parent directory	cd ..
15	mkdir	Create a new directory	mkdir OSLab
16	rmdir	Remove an empty directory	rmdir Test
17	find	Search for files and directories	find . -name "*.c"
18	locate	Quickly locate files	locate hello.c

```

chakrika@Chakrika:~$ cd ~
chakrika@Chakrika:~$ ls
FileTask  OSSP  process process.c
chakrika@Chakrika:~$ ls -l
total 28
drwxr-xr-x 2 chakrika chakrika 4096 Aug 29 08:09 FileTask
drwxr-xr-x 5 chakrika chakrika 4096 Aug 24 05:43 OSSP
-rwxr-xr-x 1 chakrika chakrika 15960 Jul 29 03:56 process
-rw-r--r-- 1 chakrika chakrika 82 Jul 29 03:56 process.c
chakrika@Chakrika:~$ ls -a
.  .bash_history  .bashrc  .config  .lessht  .motd_shown  .sudo_as_admin_successful  OSSP  process.c
.. .bash_logout  .cache  .gitconfig  .local  .profile  FileTask  process
chakrika@Chakrika:~$ cd OSSP
chakrika@Chakrika:~/OSSP$ cs ..
Command 'cs' not found, but can be installed with:
sudo apt install csound
chakrika@Chakrika:~/OSSP$ cd ..
chakrika@Chakrika:~$ mkdir Test
chakrika@Chakrika:~$ ls
FileTask  OSSP  Test  process process.c
chakrika@Chakrika:~$ rmdir Test
chakrika@Chakrika:~$ ls
FileTask  OSSP  process process.c
chakrika@Chakrika:~$ |

```

```

chakrika@Chakrika:~$ find . -name "*.c"
./process.c
./OSSP/task1.c
./OSSP/task4.c
./OSSP/Skill/Week-02/src/main.c
./OSSP/Skill/Week-02/src/builtin.c
./OSSP/Skill/Week-02/src/parser.c
./OSSP/Skill/Week-02/src/executor.c
./OSSP/Skill/Week-02/src/shell.c
./OSSP/Skill/Week-02/tests/test_parser.c
./OSSP/Skill/Week-02/tests/test_executor.c
./OSSP/task2.c
./OSSP/task3.c
./OSSP/Practical/Week-02/src/command_executor.c
./OSSP/Practical/Week-01/src/main.c
./OSSP/Practical/Week-01/src/builtin.c
./OSSP/Practical/Week-01/src/parser.c
./OSSP/Practical/Week-01/src/executor.c
./OSSP/Practical/Week-01/src/shell.c
./OSSP/Practical/Week-01/tests/test_parser.c
./OSSP/Practical/Week-01/tests/test_executor.c
./OSSP/Practical/Week-03/Task-4/process_states.c
./OSSP/Practical/Week-03/Task-3/fork_process.c
./OSSP/command_shell.c
chakrika@Chakrika:~$ locate
Command 'locate' not found, but can be installed with:
sudo apt install plocate
chakrika@Chakrika:~$ locate crode.c
Command 'locate' not found, but can be installed with:
sudo apt install plocate
chakrika@Chakrika:~$ |

```

Topic 3: File Management			
19	touch	Create a new empty file	touch file1.txt
20	cat	Display file contents	cat file1.txt
21	cp	Copy files or directories	cp a.txt b.txt
22	mv	Move or rename files	mv old.txt new.txt
23	rm	Delete a file	rm test.txt
24	head	Display first lines of a file	head hello.c
25	tail	Display last lines of a file	tail hello.c
26	more	View a file page by page	more file.txt
27	less	Scroll through a file	less hello.c
28	wc	Count lines, words and characters	wc hello.c
29	diff	Compare two files	diff a.txt b.txt
30	sort	Sort file contents	sort data.txt
31	uniq	Remove duplicate lines	uniq data.txt
32	grep	Search for text inside a file	grep main hello.c

code.c

```
#include <stdio.h>
```

```
int main() {
```

```
printf("Hello World!\n");
```

```
return 0;
```

```
}
```

```
chakrika@blue:~$ touch test.txt
chakrika@blue:~$ echo "Hello Linux" > test.txt
chakrika@blue:~$ cat test.txt
Hello Linux
chakrika@blue:~$ cp test.txt copy.txt
chakrika@blue:~$ mv copy.txt moved.txt
chakrika@blue:~$ rm moved.txt
```

```
chakrika@blue:~$ head code.c
#include <stdio.h>
int main() {
    printf("Hello World!\n");
    return 0;
}
chakrika@blue:~$ tail code.c
#include <stdio.h>
int main() {
    printf("Hello World!\n");
    return 0;
}
chakrika@blue:~$ more test.txt
chakrika@blue:~$ less test.txt
```

```
Hello Linux
```

```
~
```

```
~
```

```
~
```

```
~
```

```
~
```

```
~
```

```
~
```

```
~
```

```
(END)
```

```
chakrika@blue:~$ touch file.txt
chakrika@blue:~$ wc code.c
 5 10 73 code.c
chakrika@blue:~$ diff test.txt file.txt
1d0
chakrika@blue:~$ sort test.txt
Hello Linux
chakrika@blue:~$ uniq test.txt
Hello Linux
chakrika@blue:~$ grep Hello test.txt
Hello Linux
```

Topic 4: File Editing			
33	nano	Create or edit a file	nano hello.c
34	vi	Edit a file using Vim editor	vi hello.c

calc.c

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b;
```

```
    printf("Enter two numbers: ");
```

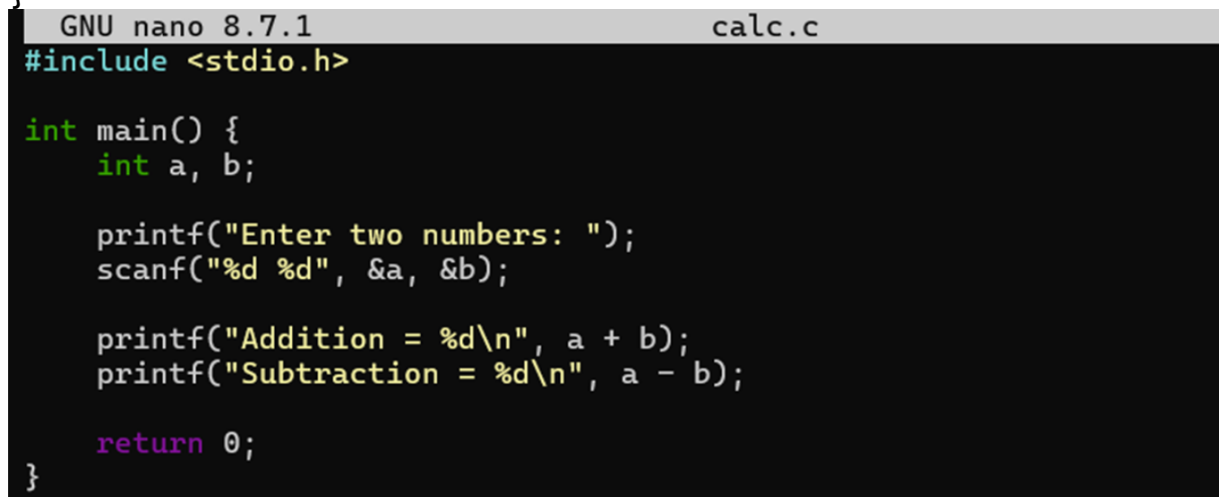
```
    scanf("%d %d", &a, &b);
```

```
    printf("Addition = %d\n", a + b);
```

```
    printf("Subtraction = %d\n", a - b);
```

```
    return 0;
```

```
}
```



```
GNU nano 8.7.1 calc.c
#include <stdio.h>

int main() {
    int a, b;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Addition = %d\n", a + b);
    printf("Subtraction = %d\n", a - b);

    return 0;
}
```

```

#include <stdio.h>

int main() {
    int a, b;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Addition = %d\n", a + b);
    printf("Subtraction = %d\n", a - b);

    return 0;
}

```

13,1 All

Topic 5: C Program Compilation and Execution			
35	gcc	Compile a C program	gcc hello.c -o hello
36	./program	Execute the compiled program	./hello
37	which	Locate the executable of a command	which gcc
38	whereis	Locate binary, source, and manual pages	whereis gcc
39	man	Display manual pages	man gcc

NAME

gcc - GNU project C and C++ compiler

SYNOPSIS

```
gcc [-c|-S|-E] [-std=standard]  
    [-g] [-pg] [-Olevel]  
    [-Wwarn...] [-Wpedantic]  
    [-Idir...] [-Ldir...]  
    [-Dmacro[=defn]...] [-Umacro]  
    [-foption...] [-mmachine-option...]  
    [-o outfile] [@file] infile...
```

Only the most useful options are listed here; see below for the remainder. g++ accepts mostly the same options as gcc.

DESCRIPTION

When you invoke GCC, it normally does preprocessing, compilation, assembly and linking. The "overall options" allow you to stop this process at an intermediate stage. For example, the -c option says not to run the linker. Then the output consists of object files output by the assembler.

Other options are passed on to one or more stages of processing. Some options control the preprocessor and others the compiler itself. Yet other options control the assembler and linker; most of these are not documented here, since you rarely need to use any of them.

Most of the command-line options that you can use with GCC are useful for C programs; when an option is only useful with another language (usually C++), the explanation says so explicitly. If the description for a particular option does not mention a source language, you can use that option with all supported languages.

Topic 6: Process Management			
40	ps	Display running processes	ps
41	ps -ef	Display all running processes	ps -ef
42	top	Monitor running processes	top
43	kill	Terminate a process	kill 1234
44	kill -9	Forcefully terminate a process	kill -9 1234

```
PID TTY          TIME CMD
385 pts/0        00:00:00 bash
554 pts/0        00:00:00 ps
```

```

UID          PID    PPID  C STIME TTY          TIME CMD
root           1         0  2 05:14 ?        00:00:00 /sbin/init
root           2         1  0 05:14 hvc0     00:00:00 /init
root           6         2  0 05:14 hvc0     00:00:00 plan9 --control-socket 7
root          46         1  0 05:14 ?        00:00:00 /usr/lib/systemd/systemd
systemd+      63         1  0 05:14 ?        00:00:00 /usr/lib/systemd/systemd
root          88         1  1 05:14 ?        00:00:00 /usr/lib/systemd/systemd
root         154         1  0 05:14 ?        00:00:00 /bin/sh /usr/lib/systemd
root         155         1  0 05:14 ?        00:00:00 /usr/sbin/cron -f -P
message+     156         1  0 05:14 ?        00:00:00 @dbus-daemon --system --
root         162         1  0 05:14 ?        00:00:00 /usr/bin/python3 /usr/bi
root         171         1  0 05:14 ?        00:00:00 /usr/lib/systemd/systemd
root         174         1  0 05:14 ?        00:00:00 /usr/libexec/wsl-pro-ser
syslog       206         1  0 05:14 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root        260         1  0 05:14 tty1     00:00:00 /usr/sbin/agetty --nores
_chrony      262       154  0 05:14 ?        00:00:00 /usr/sbin/chronyd -n -F
_chrony      271       262  0 05:14 ?        00:00:00 /usr/sbin/chronyd -n -F
root        345         1  0 05:14 ?        00:00:00 /usr/bin/python3 /usr/sh
root        380         2  0 05:14 ?        00:00:00 /init
root        383       380  0 05:14 ?        00:00:00 /init

1:Def - 05:15:23 up 1 min, 1 user, load average: 0.09, 0.04, 0.01
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7552.6 total, 6202.5 free, 561.5 used, 939.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 6991.2 avail Mem

1:Def - 05:15:23 up 1 min, 1 user, load average: 0.09, 0.04, 0.01
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7552.6 total, 6202.5 free, 561.5 used, 939.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 6991.2 avail Mem

1  PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM    TIME+  COMMAND
  1 root       20   0   24224  15500  11664 S   0.0   0.2   0:04.27 systemd
  2 root       20   0   3180   2204   2072 S   0.0   0.0   0:00.18 init-systemd(Ub
  6 root       20   0   3180   2044   1940 S   0.0   0.0   0:00.00 init
 46 root       19  -1   42204  16568  15308 S   0.0   0.2   0:00.18 systemd-journal
1:Def - 05:15:35 up 1 min, 1 user, load average: 0.07, 0.04, 0.01
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7552.6 total, 6201.2 free, 562.7 used, 939.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 6990.0 avail Mem

1  PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM    TIME+  COMMAND
  1 root       20   0   24224  15500  11664 S   0.0   0.2   0:00.79 systemd
  2 root       20   0   3180   2204   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
  6 root       20   0   3180   2044   1940 S   0.0   0.0   0:00.00 init
 46 root       19  -1   42204  16572  15308 S   0.0   0.2   0:00.18 systemd-journal
 63 systemd+  20   0   22540  14624  12004 S   0.0   0.2   0:00.09 systemd-resolve

2  PID PPID    %CPU  %MEM  PR  NI S  VIRT   RES  UID COMMAND
 577 385    0:00.04 0.0  0.1 20  0 R   8164   6088 1000 top
 469 386    0:00.03 0.0  0.1 20  0 S   6136   5492 1000 bash
 441 439    0:00.00 0.0  0.1 20  0 S   22916  4080 1000 (sd-pam)

```

Topic 7: Job Control			
45	jobs	Display background jobs	jobs
46	bg	Resume a job in the background	bg %1
47	fg	Bring a job to the foreground	fg %1
48	&	Run a process in the background	./hello &

```

chakrika@blue:~$ sleep 100 &
[1] 1311
chakrika@blue:~$ jobs
[1]+  Running                  sleep 100 &
chakrika@blue:~$ bg %1
-bash: bg: job 1 already in background
chakrika@blue:~$ fg %1
sleep 100
chakrika@blue:~$ ./code &
[1] 1360
Hello World!
[1]+  Done                      ./code

```

Topic 8: File Permissions and Links			
49	chmod	Change file permissions	chmod +x script.sh
50	chmod 755	Set read, write, and execute permissions	chmod 755 script.sh
51	ln	Create a hard link	ln file1 file2
52	ln -s	Create a symbolic link	ln -s file1 shortcut

```
total 220
-rw-r--r-- 1 amrutha amrutha 135549 Apr 23 17:07 InRelease
drwxr-xr-x 11 amrutha amrutha 4096 Aug 3 06:03 OSSP
-rwxr-xr-x 1 amrutha amrutha 16056 Aug 8 09:11 calc
-rw-r--r-- 1 amrutha amrutha 208 Aug 8 09:09 calc.c
-rwxr-xr-x 1 amrutha amrutha 15952 Aug 5 06:29 code
-rw-r--r-- 1 amrutha amrutha 73 Aug 8 09:04 code.c
-rw-r--r-- 1 amrutha amrutha 12 Aug 8 08:39 demo.txt
-rw-r--r-- 1 amrutha amrutha 0 Aug 8 08:59 file.txt
-rw-r--r-- 2 amrutha amrutha 12 Aug 8 08:41 hardlink.txt
-rwxr-xr-x 1 amrutha amrutha 15952 Jul 28 06:28 process
-rw-r--r-- 1 amrutha amrutha 80 Jul 28 06:28 process.c
drwxr-xr-x 6 amrutha amrutha 4096 Jul 28 09:49 pytorch-env
lrwxrwxrwx 1 amrutha amrutha 8 Aug 10 07:50 shortcut.txt -> test.txt
-rwxr-xr-x 1 amrutha amrutha 15 Aug 10 07:50 test.sh
-rw-r--r-- 2 amrutha amrutha 12 Aug 8 08:41 test.txt
```

Topic 9: Disk and Memory Information			
53	df -h	Display disk space usage	df -h
54	du -sh	Display directory size	du -sh .
55	free -h	Display memory usage (Linux)	free -h

```
Filesystem      Size  Used Avail Use% Mounted on
none            3.7G  0 3.7G  0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.7G  4.0K 3.7G  1% /mnt/wsl
drivers          476G  207G  270G  44% /usr/lib/wsl/drivers
/dev/sdd        1007G   9.4G  947G  1% /
none            3.7G   32K 3.7G  1% /mnt/wslg
none            3.7G  0 3.7G  0% /usr/lib/wsl/lib
rootfs          3.7G  2.8M 3.7G  1% /init
none            3.7G  528K 3.7G  1% /run
none            3.7G  0 3.7G  0% /run/lock
none            3.7G  0 3.7G  0% /run/shm
none            3.7G  104K 3.7G  1% /mnt/wslg/versions.txt
none            3.7G  104K 3.7G  1% /mnt/wslg/doc
C:\              476G  207G  270G  44% /mnt/c
none            1.0M  0 1.0M  0% /run/credentials/systemd-journald.service
tmpfs           3.7G  0 3.7G  0% /tmp
none            1.0M  0 1.0M  0% /run/credentials/systemd-resolved.service
none            1.0M  0 1.0M  0% /run/credentials/getty@tty1.service
tmpfs           756M  12K 756M  1% /run/user/1000
```

Topic 10: User and Environment Management

56	id	Display user and group IDs	id
57	env	Display environment variables	env
58	export	Set an environment variable	export PATH=\$PATH:.
59	alias	Create a command shortcut	alias ll='ls -l'

```
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=blue
PWD=/home/amrutha
LOGNAME=amrutha
HOME=/home/amrutha
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/326_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:
:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;3
1:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:
*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.taz=01;31:*.deb=01;31:*.rpm=01;31:
*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:
*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=0
1;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg
=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:
*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35
:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:
*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;3
6:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:*.bak=00;90:*.old=00;90:*.
orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpgk-dist=00;90:*.dpgk-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.u
cf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=amrutha
DISPLAY=:0

total 220
-rw-r--r-- 1 amrutha amrutha 135549 Apr 23 17:07 InRelease
drwxr-xr-x 11 amrutha amrutha 4096 Aug 3 06:03 OSSP
-rwxr-xr-x 1 amrutha amrutha 16056 Aug 8 09:11 calc
-rw-r--r-- 1 amrutha amrutha 208 Aug 8 09:09 calc.c
-rwxr-xr-x 1 amrutha amrutha 15952 Aug 5 06:29 code
-rw-r--r-- 1 amrutha amrutha 73 Aug 8 09:04 code.c
-rw-r--r-- 1 amrutha amrutha 12 Aug 8 08:39 demo.txt
-rw-r--r-- 1 amrutha amrutha 0 Aug 8 08:59 file.txt
-rw-r--r-- 2 amrutha amrutha 12 Aug 8 08:41 hardlink.txt
-rwxr-xr-x 1 amrutha amrutha 15952 Jul 28 06:28 process
-rw-r--r-- 1 amrutha amrutha 80 Jul 28 06:28 process.c
drwxr-xr-x 6 amrutha amrutha 4096 Jul 28 09:49 pytorch-env
lrwxrwxrwx 1 amrutha amrutha 8 Aug 10 07:50 shortcut.txt -> test.txt
-rwxr-xr-x 1 amrutha amrutha 15 Aug 10 07:50 test.sh
-rw-r--r-- 2 amrutha amrutha 12 Aug 8 08:41 test.txt
```